

Automated Essay Scoring (AES) using English Essay Question

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Abstract—Automated Essay Scoring (AES) is a one of the important research areas in educational technology. Research about AES is started in the early 1960s and keep growing in development along with the advances of the computing technology. AES is more widely use by the educational system for assessment and classroom activities, it being use in all layers of education. The potential for Automated Essay Scoring is being used widely becomes a reality as more and more classroom activities, assessment, and test preparation materials are supplied online. These essay questions might include everything from arithmetic, where students are asked to explain how they arrived at their answers, to science, where they might be asked to define words or explain experiments, or history, where they must ask to explain or discuss an event. In this study we try to design the AES using Natural Language Processing (NLP) and machine learning techniques to evaluate the student's answer by comparing it with the answer scheme. It is not simply a string-matching program and need a proper system to process the students answer. We propose the 4 main process to the system that is Spelling Checking, Pre-Processing, Latent Semantic Analysis and Thesaurus Checking. All this process is combined and analyze to become a good AES system. Each of this process mean give somethings important to the system flow on how the essay going to be evaluate. The system has been done some experiment and the system get the average correlation between system scores and human appraiser scores is only 70%. However, there needs to be more research on the future use of this Automatic Essay Scoring system.

Keywords—Automated Essay Scoring; Natural language processing; Corpus; Latent Semantic Analysis; Wordnet

I. INTRODUCTION

The educational community is growing relentlessly with more and more students, curriculum, and exams. The growing community raises the need for a scoring system that eases the burden of scoring many exams and at the same time guarantees the fairness of the scoring process [1].

As more and more assessments, test preparation and teaching materials are sent online, it is possible for automatic examination of test answers or quizzes to become a reality and so is the need in this technological age. To this date, research in automatic scoring has focused on scoring essay questions as opposed to multiple choice questions that is typically used for homework assignments, classroom tests, and chapter final review questions.

These kinds of questions can cover anything from math, where students are asked to explain how they get their answers, to science, where they might be asked to define terms, explain experiments, or read comprehension questions, to history, where they might be asked to explain or discuss an event. The Automatic Essay Scoring System (AES) evaluates students' answers by comparing them with model answers. The AES usually uses Natural Language Processing and machine learning techniques to evaluate essays written automatically [2].

In this sophisticated age, the assessment of School Questions in Malaysia focuses on essay-length questions as opposed to multiple choice questions. This is because the question in the form of (HOTS) Higher Order Thinking Skills which is a question that requires high-level thinking skills is often asked in long questions of essays in schools throughout Malaysia. In addition, the AES system has not been able to compare the rate of equality with the human assessment for fact-based questions in the assessment in schools in Malaysia. This is since students' answers from exam questions in schools in Malaysia have never been automatically evaluated by the AES system.

According to Jason Brownlee [3], NLP is automatic manipulation of the natural properties of a language such as speech or text in the software application. The study of the process of the natural nature of language has taken place over the past 50 years and it continues to grow from the field of linguistics with the advancement of computers that are becoming more and more sophisticated. According to the Dictionary corpus is a collection of papers on a specific matter or group of materials for study. By other purposes corpus is a genuine collection of text and audio collected and sorted into data sets. In the NLP a corpus contains text and speech data that can be used to train "artificial intelligence" and "machine learning" systems. In a software study into machine-learned conversation patterns using "machine learning" from the corpus of transcript dialogues has been used to generate various chatbots that speak multiple languages and sublanguages including English, France, and Arabic. Among the most famous AES systems applied to English are e-rater [4], C-rater [5], Oxford-UCLES, Intelligent Essay Assessor™ (IEA), IndusMarker and Text-to-Text system.

e-rater use NLP to identify the characteristics of faculty essays in sample collections and store them relevant in a large database. When the e-rater evaluates a new essay, it compares the characteristics of the latest essay with the characteristics of the essay in the database to set the score [6].

C-rater is a short answer scoring engine, developed by ETS Technologies, designed for responses of answers built into questions that measure the understanding of content material. C-rater differs from the essay scoring system in several basic and important ways that prioritizes its main task - recognizing the paraphrase or equivalent meaning [7].

Unlike other AES systems, the IEA's™ focus is more on content-related features rather than on writing forms. However, this does not mean that the IEA™ does not provide feedback on formal aspects (for example, grammar and punctuation marks) in the essay. In other words, although the system uses an LSA-based approach to evaluate primarily the quality of the essay content, it also includes scoring and feedback on the grammar, style, and mechanics of answer writing [8].

Based on the title of the study, the purpose of this research is to achieve the objective of designing an Automatic Essay Scoring (AES) system suitable for questions based on the fact-based high secondary examination in Malaysia. and to compare the rates of similarity between the assessment of the AES System and the human evaluation. In this study we try to design the Automated Essay Scoring (AES) using Natural Language Processing (NLP) and machine learning techniques

II. METHODOLOGY

Question and Answer Criteria

First for all is that the questions that is taken must be from the Latest Assessment of schools throughout Malaysia. This study focuses on the assessment of high secondary students i.e., students from form 4 to form 5. In addition, the questions in the form of this essay consist of fact-based questions, in which they may be required to explain the experiment or define the term. For example, a physics subject question that asks a question that requires understanding the facts of a concept as shown in Fig. 1. Finally, the answers of the students who want to be used must be written in English only.

3. Jadual 3 menunjukkan dua buah alat pengukuran yang digunakan untuk mengukur diameter sekeping duit syiling.
Table 3 shows two measuring instruments used to measure the diameter of a coin.

Ciri-ciri Characteristics	Alat pengukuran A Measuring Instrument A	Alat pengukuran B Measuring Instrument B
Senggatan terkecil Smallest division	0.01 cm	0.01 mm
Julat / Range	0.00 cm – 15.00 cm	0.00 mm – 25.00 mm
Kepekaan / Sensitivity	Tinggi / High	Sangat tinggi / Very high

- (a) Berdasarkan Jadual 3, bandingkan senggatan terkecil, julat serta kepekaan kedua-dua alat pengukuran tersebut. Seterusnya, nyatakan hubungan kait antara senggatan terkecil dengan kepekaan alat pengukuran tersebut.
Based on Table 3, compare the smallest scale, range and sensitivity for both measuring instruments. Then, state the relationship between the smallest division and the sensitivity of the measuring instruments. [4 markah / 4 marks]
- (b) Terangkan mengapa pengukuran diameter duit syiling diulang beberapa kali.
Explain why the measurement of the coin's diameter was repeated a few times. [4 markah / 4 marks]

Fig. 1. Examples of essay questions for the subject of Physics

Experiment Design and Procedure

The design of this study was carried out in a simulated manner. The approach used in this study is software and does not involve any hardware. To reshape the automatic marking system will use the programming language "Python". The use of Python is because Natural Language Processing [9] with Python provides a practical introduction to programming for language processing. With NLTK that is the main platform for building Python programs to relate to human language data. It provides an easy-to-use interface for more than 50 corrugation and lexical sources [10] such as WordNet [11], along with a group of text processing libraries for classification, tokenization, marking, elaboration, and semantic reasoning, wrapping to strengthen the NLP library industry and active discussion forums.

Fig. 2 shows a flowchart to build an Automatic Essay Scoring system by using Jupyter Notebook, a web-based interactive computing platform. Jupyter notebook is an open-source IDE that is used to build Jupyter documents that can be created and mutual with live codes. It contains 4 main processes, namely Spelling Checking, Preprocessing, Latent Semantic Analysis and Thesaurus Checking.

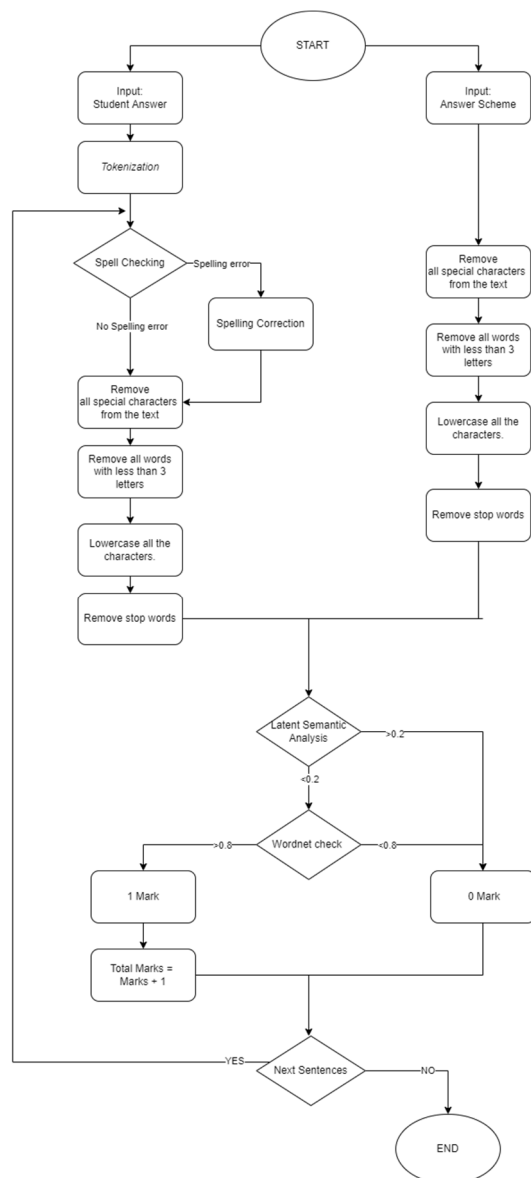


Fig. 2. Flow Chart of AES

Based on the flowchart above we can see that there are two inputs which are student essay answers and answer schemes. Student essay answers are expected to be given in the form of essays so that the input can be processed in this Automatic Essay Scoring system. As for the input of the answer scheme, input is asked to be given in the form of a point because most of the answer schemes of a question or test are given sentences that carry a certain score.

Design of the System

The first process when getting an essay answer from a student is to check the spelling of each word in a paragraph. If there is a spelling error in the essay the system will automatically make spelling corrections to the answer. This is to ensure that the

answers can be checked accurately and in detail. But if the spelling error is too big that the system does not understand a word in a sentence then the spelling check will not be done and the score for the sentence will not be given.

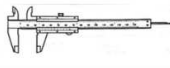
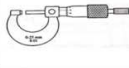
The most important part of any machine learning algorithm is data preprocessing step. Preprocessing this data is done is to remove non-essential word interference to ensure the keyword in each given answer. More clutter found in the data leads to a lack of accuracy of the model. Data preprocessing consists of four types of processing on data for example Remove all special characters from the text, remove all words with less than 3 letters, lowercase the letters of all words, Remove Stopwords.

Latent Semantic Analysis is an efficient way to analyze the text and find hidden topics by understanding the context of the text. Latent Semantic Analysis (LSA) is used to find hidden topics represented by documents or text. Using the concept of Singular Value Decomposition (SVD) and cost equality (Cos similarity) as the mainstay of this analysis. This hidden topic is then used to compile the same document. LSA is an unsupervised algorithm and therefore we do not know the exact topic of the document. For the Latent Semantic Analysis Process two sentences taken from the student's answer and the answer scheme are compared to get the value of the topic as close as possible to ensure that these two sentences are in the same topic.

Thesaurus is a source that collects a list of words related to their meaning that are sorted according to a specific system or processing. This thesaurus examination process is the final process before the sentence can be scored or not after the Latent Semantic Analysis has ensured that the student's answer sentences and the answer sentences from the answer scheme are in the context of the same topic. This examination is a certainty so that the answer given by this student has the same meaning or the hamper is like the schematic answer. For this Thesaurus Inspection process the corpus used is Wordnet.

Graphical User Interface

Designing the visual composition and behavior of the graphical user interface is an important part of software application programming such as the Automatic Essay Scoring (AES) system in the field of human-computer interaction. The goal is to increase the efficiency and ease of use for the logical design of the created program. In the graphical user interface of the Automatic Essay Scoring system, there are several media elements such as text and diagrams or tables that are always given in quiz questions or exams to make it easier for users to answer the questions given. Fig. 3 is an example of a graphical user interface designed in the Essay Scoring system.

	Measuring Instrument A	Measuring Instrument B
Characteristics		
Smallest Division	0.01cm	0.01mm
Range	0.00cm – 15.00cm	0.00mm – 25.00mm
Sensitivity	High	Very High

Dear Student

a) Based on Table 3, compare the smallest scale, range and sensitivity for both measuring instrument. Then state the relationship between the smallest division and the sensitivity of the measuring

Question a: Answer Here

b) Explain why the measurement of the coin's diameter was repeated a few times

Question b: Answer Here

Send Answer

Fig. 3. Graphical User Interface for AES

Data Collection

The data needed to achieve the second objective which is to compare the rates of equality between the assessment of the Automatic Essay Scoring System and the human assessment is the student's own answer to the given question. As suggested by the student data, the student's answer should be done online on a computer or other electronic device so that it can meet the criteria of the Automatic Essay Scoring System.

But due to the lack of time to conduct this study, the student's answer data and the scheme are taken manually due to the difficulty of conducting experiments on this system face-to-face. 10 student answers were taken to conduct this experiment. Questions are taken from the question of the final chapter of the training book of the Form 4 Physics subjects. 1 The set of answer schemes is also taken to be an automatic evaluation guide. This set of questions and set of answer schemes are proposed by the teacher herself as it has been evaluated by the teacher herself. This is the best way that can be done to perfect the second objective with this very short time. Fig. 4 shows the set of student answers taken manually.

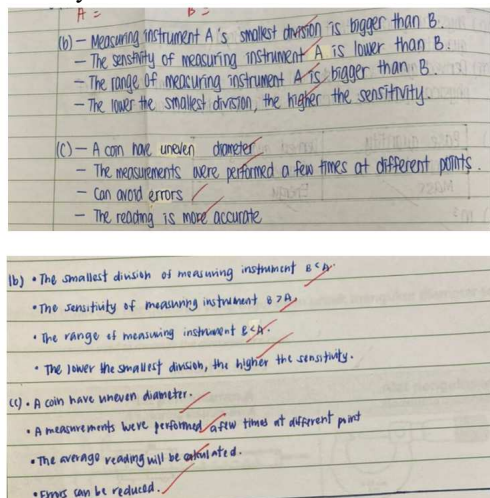


Fig. 4. Student answers

III. RESULTS AND DISCUSSION

Spelling Checking Analysis

In this analysis the spelling checking process can be done when given an essay answer. The set-set of essay answers is taken and performed the Spell-Checking process to determine if it is appropriate for this spell-checking process in this Essay Scoring system. Using the Speller function from the autocorrect library, it will check the spelling of the given input. Fig. 5 shows a set of answers as inputs that have the correct spelling of each word and there is no change when this spell-checking process is performed.

```
from autocorrect import Speller
from nltk.tokenize import sent_tokenize

spell = Speller(lang='en')

answer_text = input("Enter answer : ")
corrected_answer_text = spell(answer_text)

tokenize_answer_text = str(sent_tokenize(corrected_answer_text))
print(tokenize_answer_text)

Enter answer : The Smallest division of measuring instrument B < A.
['The Smallest division of measuring instrument B < A. ']
```

Fig. 5 Spelling Checking Process

Latent Semantic Analysis

Once the student's answer has gone through the Spell Checking and Preprocessing process, both input and output will be compared to the topic equation with the Latent Semantic Analysis method [12,13]. With this analysis it can be found out that both input and output have a topic similarity or vice versa. Fig. 6 shows that both the student's answer and the answer scheme are compared and get a value of 2.44949 as the value of the topic for both topics. With the same value this makes these two sentences be in the same topic.

```
import pandas as pd

topic_encoded_df = pd.DataFrame(1sa,columns = ["topics_1"])
topic_encoded_df["parts"] = parts
display(topic_encoded_df[["parts","topics_1"]])

topic_encoded_df.at[0,'topics_1']
```

	parts	topics_1
0	["The smallest division of measuring instrume...	2.44949
1	["The smallest division of measuring instrume...	2.44949

Fig. 6. Latent Semantic Analysis Process

Thesaurus Checking Process

This analysis is the final analysis of the Automatic Essay Scoring system, and it is a determinant for the answer students will score 1 or 0. This analysis wanted to see whether comparing keywords for both student answers and answer schemes provided an important role in the system. In addition, does it effectively give a synonym value using the Wu and Palmer methods in the comparison between these two words [9].

Fig. 7 shows the division keywords and instruments taken from the student's answers and the answer scheme and it looks for the synonymous values of the two keywords. Both keywords are the same and they give the same synonym value which is 1 for the word division and instrument. With a synonym value that exceeds the value of 0.8 as set out in the system for this Thesaurus Examination

process, it is eligible for one of the sentences to which this student answers to get 1 mark for the compared sentence only.

```
for i in range(0,len (tag_scheme)):
    if tag_scheme[i][1] == 'NN' or tag_scheme[i][1] == 'JJR':
        print(tag_scheme[i][0])
    if tag_answer[i][1] == 'NN' or tag_answer[i][1] == 'JJR':
        syn1 = wordnet.synsets(tag_answer[i][0])[0]
        syn2 = wordnet.synsets(tag_scheme[i][0])[0]

        print ("Answer keyword : ", syn1.name())
        print ("Scheme keyword : ", syn2.name())
```

```
division
Answer keyword : division.n.01
Scheme keyword : division.n.01
1.0
instrument
Answer keyword : instrument.n.01
Scheme keyword : instrument.n.01
1.0
```

Fig. 7. Thesaurus Checking Process

Teacher Verification

Because this system is a new system, Teacher verification is the last step that before a student's score is finalized. This process is an additional process for the early stages of the study to be implemented so that teachers can check and ensure that the system provides fair marks for all students. Fig. 8 shows students' answers, and their scores are inserted into Microsoft Excel can be a document that the teacher can see and evaluate the student's answers and the marks given by the system.

No.	Student Answer	Suggest Mark
1	1. Measuring instrument A's smallest division is bigger than B. The sensitivity of measuring instrument A is lower than B. The range of measuring instrument A is bigger than B. The lower the smallest division, the higher the sensitivity.	3
2	2. The smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	4
3	3. Micrometer Screw Gauge has the smaller smallest division the Vernier Calliper. Micrometer Screw Gauge has longer range than Vernier Calliper. The smaller the smallest division, the higher the sensitivity.	1
4	4. The Smallest Division of measuring instrument A is bigger than B. The range of measuring instrument A is bigger than B. The sensitivity of measuring instrument A is lower than B. The bigger the smallest division, the lower the sensitivity.	1
5	5. The Smallest scale of measuring instrument B-A. The range of measuring instrument B-A. The sensitivity of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	3
6	6. The smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	4
7	7. The smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	4
8	8. The Smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	2
9	9. The smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	2
10	10. The smallest division of measuring instrument B-A. The sensitivity of measuring instrument B-A. The range of measuring instrument B-A. The lower the smallest division, the higher the sensitivity.	4

Fig. 8. Teacher Verification in Microsoft Excel

The Rates of Similarity

The analysis compares the rate of similarity between the assessment of the automatic scoring system and the human evaluation to measure the performance of the system that has been developed, carried out 2 experiments and tests, the first of which is the Measurement of a human appraiser. In this experiment, it measured the average correlation

between system scores and scores of human assessors. This can also be seen as the accuracy of the system, which is a quantitative assessment of the system and Fig. 9 shows the average correlation between the system score and the scores of human appraisers. The second is Processing time. In this experiment, it measured the average time required by the system required to evaluate the student's answer.

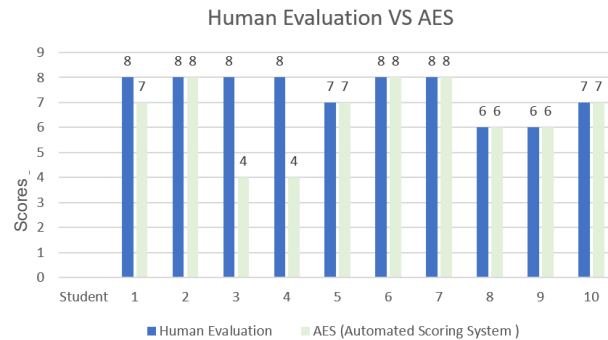


Fig. 9 Bar Chart Human Evaluation VS AES

IV. CONCLUSION

The two objectives in this study which is to design an Automatic Essay Scoring (AES) system that is suitable for fact-based questions in Malaysia and comparing the rate of assessment similarities between the AES system and human rates. For this study both objectives were achieved successfully.

For the objective of designing an Automatic Essay Scoring (AES) system suitable for fact-based questions in Malaysia, a good system can be produced as planned. This Automatic Essay Scoring system is built with a process that has been analyzed and combined into a good system.

For the second objective, comparing the rate of assessment equality between the AES system and the human rate has been achieved with corrugation is 70%. Although it did not meet the 86% quorum expectations, the objectives of this study were well achieved.

For upcoming study about this study is to design an Automatic Essay Scoring (AES) system that is suitable for linguistic questions and requires a high understanding of answers. This is because there are also essay questions asked in English subjects in the assessment in school schools especially in secondary schools. Using other techniques for this Automatic Essay Scoring system so that it can be compared corrugation to each other.

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